

L5

CLASS X - SCIENCE

METALS AND NON-METALS

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Josh
Meter

The ores of many metals are oxides, because oxygen is very reactive and abundant in the earth's crust.

K
Na
Ca
Mg
Al
Zn
Fe
Pb
H
Cu
Hg
Ag
Au
Pt

VERY REACTIVE METALS

- They are never found in free state in nature.
- They always occur in combined state as their oxides, carbonates or sulphides.

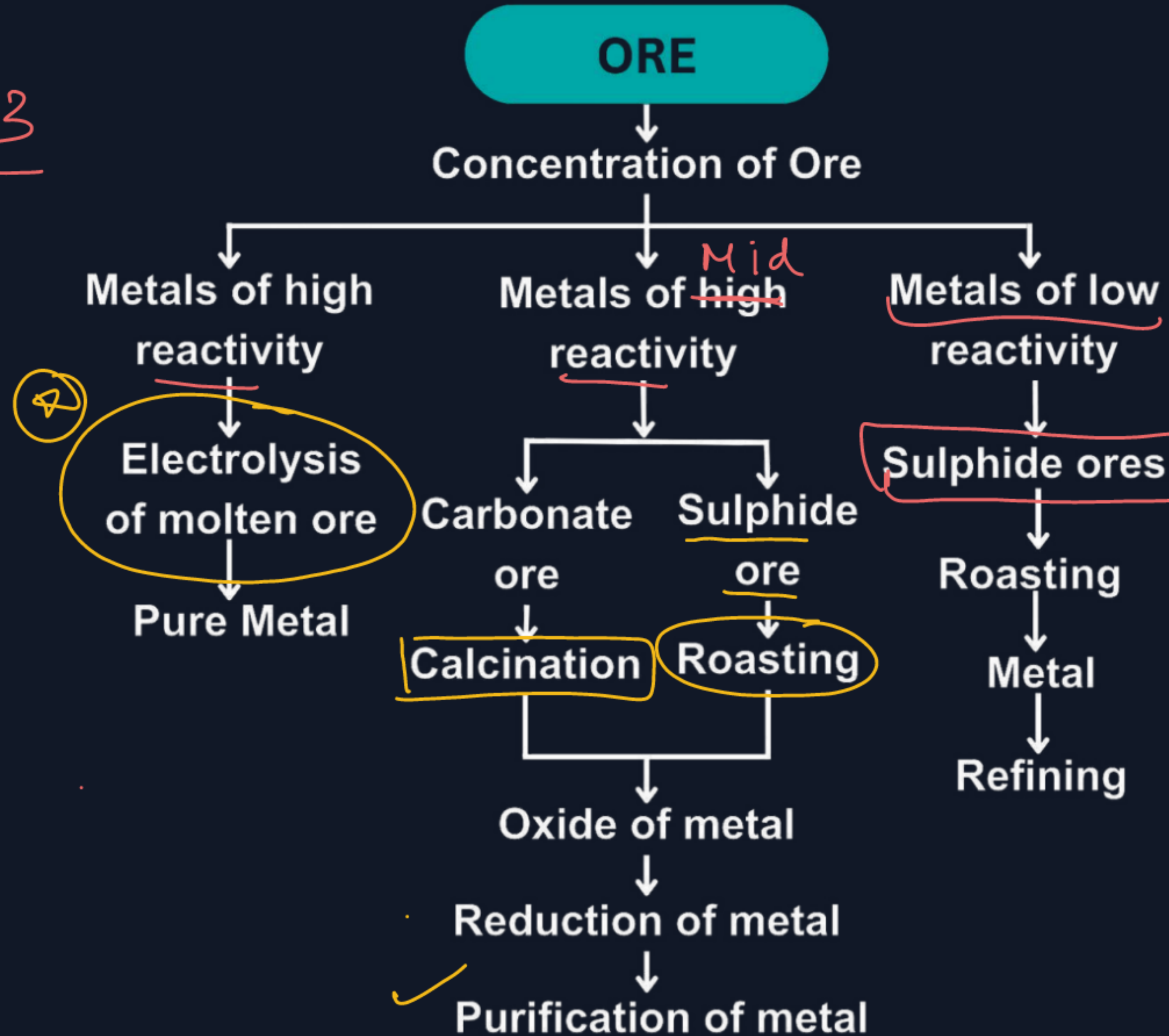
MODERATELY REACTIVE METALS

- They are found in the Earth's crust mainly as oxides, sulphides and carbonates. Zinc → as ZnS (zinc blende), ZnCO₃ (calamine), Iron → as Fe₂O₃ (haematite), Lead → as PbS (galena)

LEAST REACTIVE METALS.

- They are found in free state (native state) as well as in combined state.
- Examples: Copper, Silver, Mercury → found both free and combined (as oxides/sulphides).
- Gold, Platinum → found in free (native) state.

$ZnCO_3$



Steps in Metallurgy

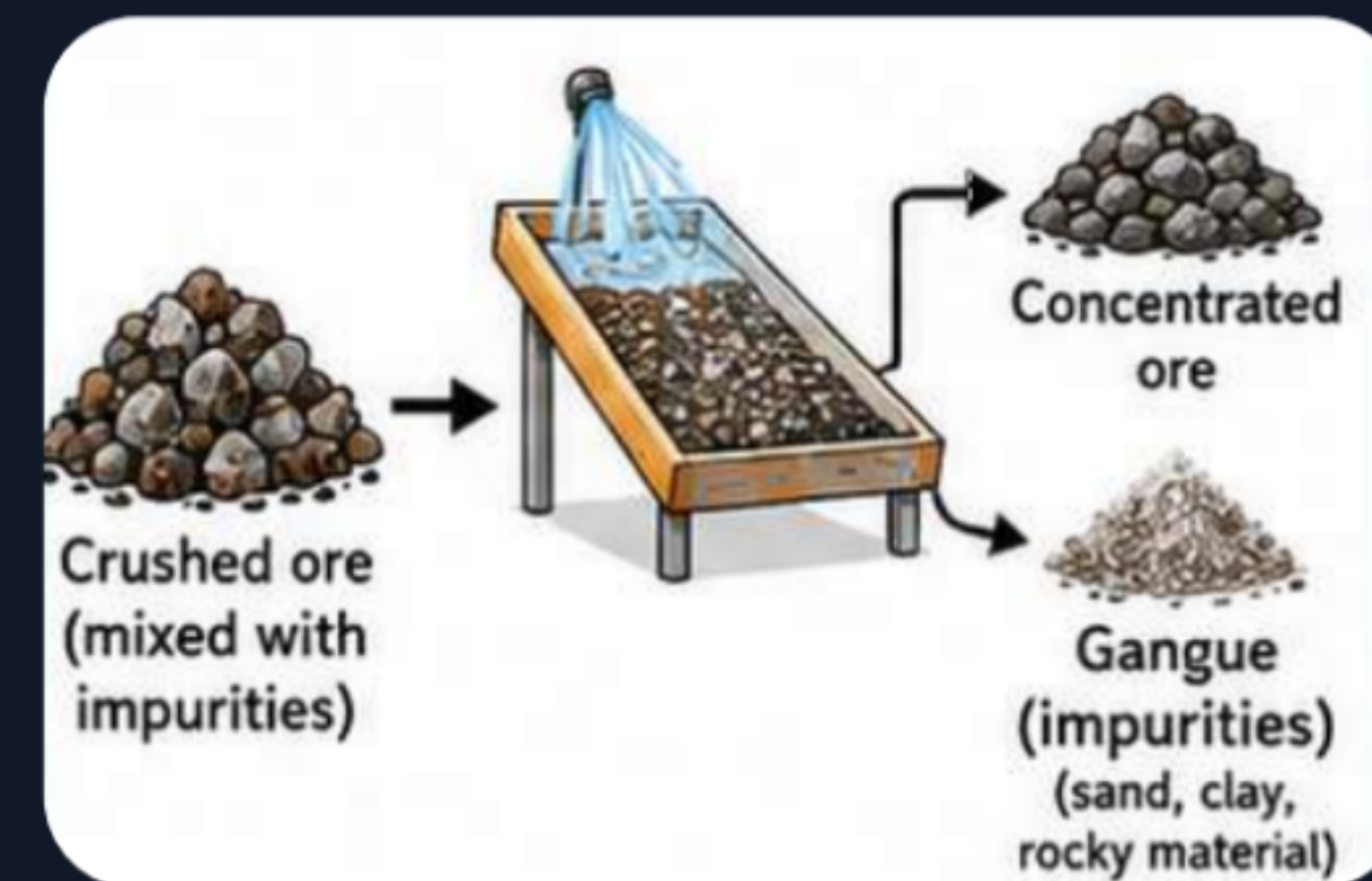
✓ Step 1: Crushing and Grinding of Ore

- First step of the metallurgical process.
- Ore is crushed and ground into smaller pieces/powder.
- Purpose → makes extraction of metal easier and more efficient.



✓ Step 2: Concentration of Ore (Enrichment)

- Process of removing gangue (impurities) like sand, clay, and rocky materials from ore.
- Increases the percentage of metal in the ore.
- Done by various physical/chemical methods depending on the ore and impurity.



Step 3: Extraction of Metals

- Process of obtaining metal from concentrated ore.
- Achieved by reduction (removal of oxygen or non-metallic element).
- Method depends on the reactivity of the metal:
- **High reactivity metals (K, Na, Ca, Mg, Al):** extracted by electrolysis.
- **Medium reactivity metals (Zn, Fe, Pb, etc.):** extracted by reduction with carbon/CO, or displacement.
- **Low reactivity metals (Au, Ag, Cu, etc.):** found in native state or extracted by mild reducing agents.



1 High reactivity metals K, Na, Ca, Mg, Al



3 Low reactivity metals Au, Ag, Cu, etc.



2 Medium reactivity metals Zn, Fe, Pb, etc.





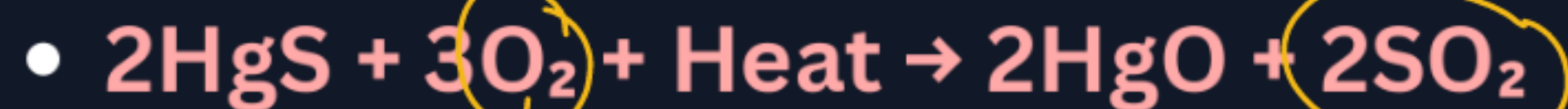
EXTRACTION OF METALS

1. Extracting metals of Low Reactivity

- Metals low in activity series → very unreactive.
- Their oxides can be reduced to metals by heating alone.

Example 1: Mercury from Cinnabar (HgS)

Step 1: Heating cinnabar in air → forms mercuric oxide (HgO)



Step 2: Heating mercuric oxide → decomposes to mercury.

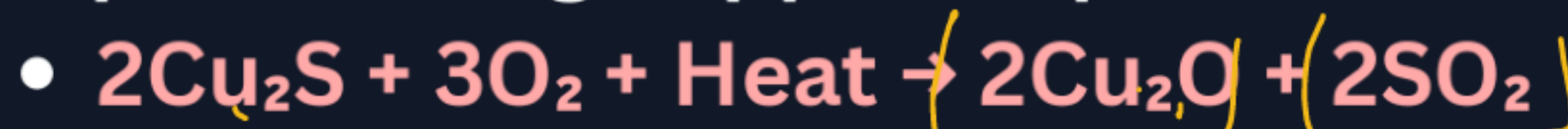


80	200.59
Hg	
Mercury	

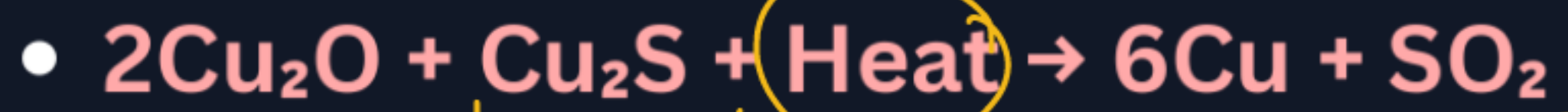
1. Extracting metals of Low Reactivity

Example 2: Copper from Copper Glance (Cu_2S)

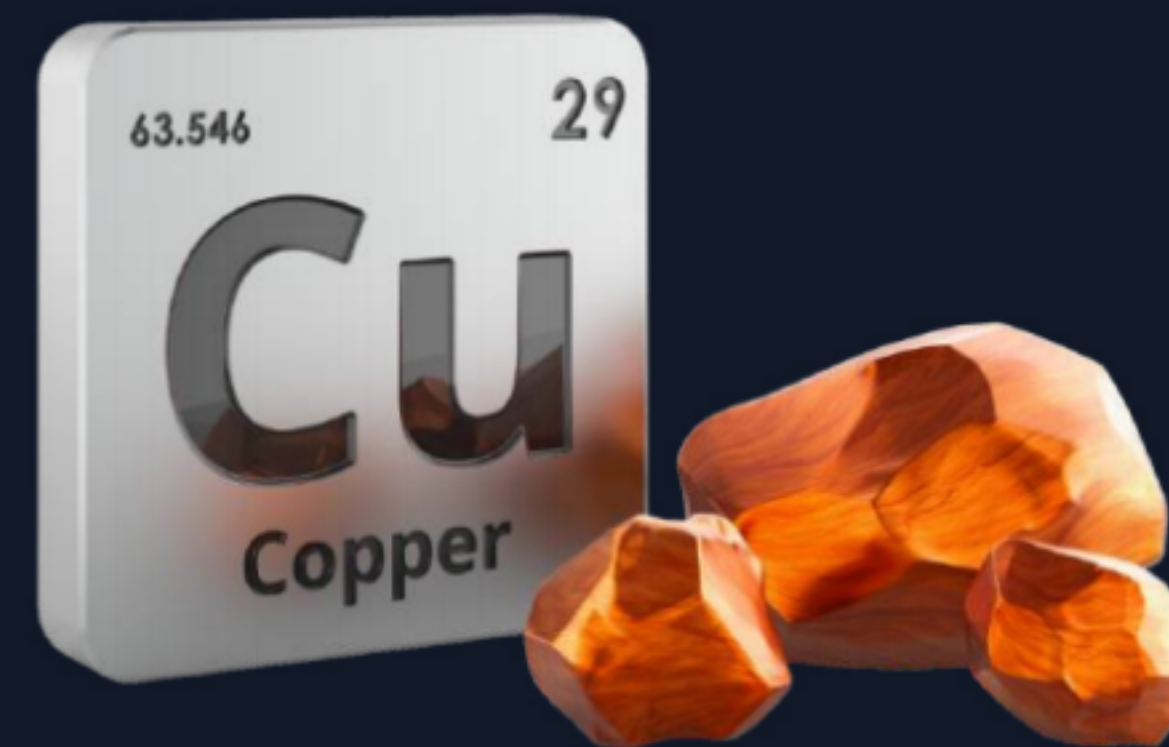
Step 1: Heating copper sulphide in air $\rightarrow$ forms copper(I) oxide (Cu_2O).



Step 2: Copper(I) oxide reacts with more copper sulphide $\rightarrow$ gives copper metal



Q. Describe the steps involved in the extraction of copper from its sulphide ore. Write the chemical equations involved.



Check yourself

Question: During extraction of copper from Cu_2S , the ore is heated in air. What is the main purpose of heating?

- ✓ 1. To convert sulfide into oxide.
2. To dissolve copper.
3. To increase impurities
4. To produce hydrogen.



Check yourself

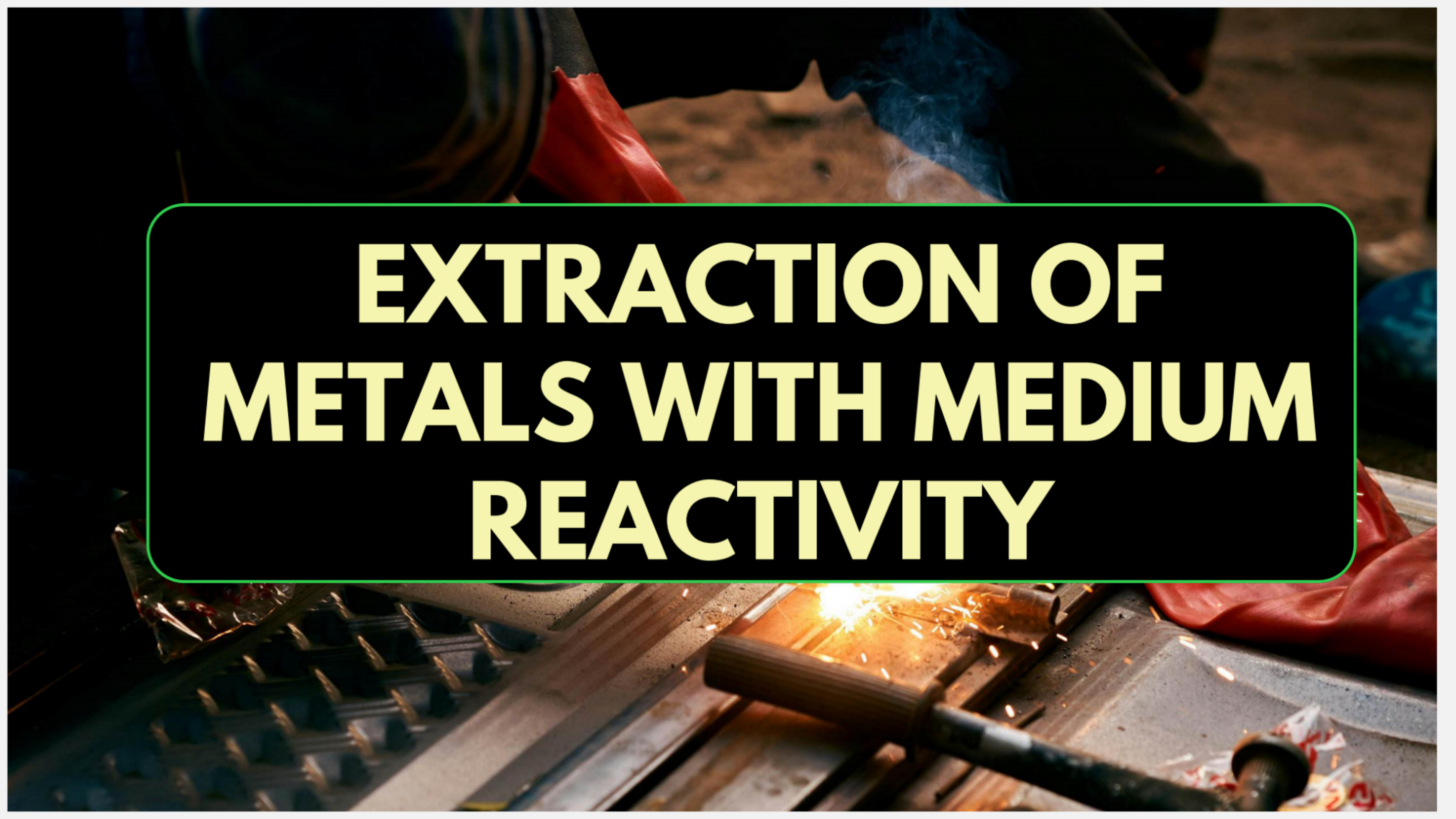
QUES: Name an ore of mercury and state the form in which mercury is found in it. Write the chemical equations for the extraction process.



Check yourself

QUES: Cinnabar is an ore of a metal 'X'. When this ore is heated in air, it is first converted into oxide of 'X' (XO) and then reduced to metal 'X' on further heating. Identify metal 'X' and write chemical equations for the reactions that occur in the above processes.



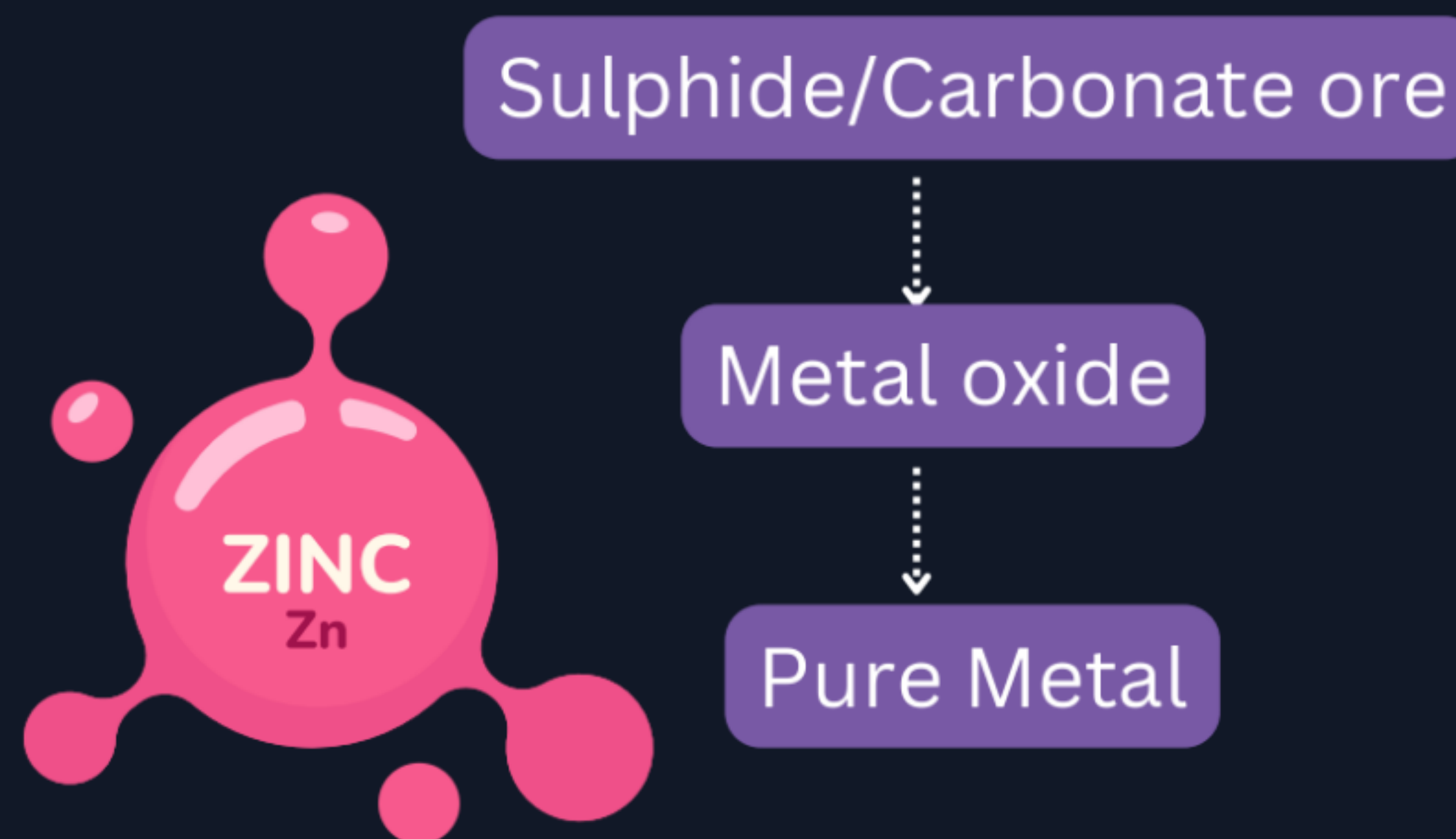


EXTRACTION OF METALS WITH MEDIUM REACTIVITY

Extraction of Medium reactive Metals

- Metals like **iron (Fe)**, **zinc (Zn)**, **lead (Pb)** are moderately reactive.
- Found as sulphides (MS) or carbonates (MCO_3) in nature.
- Easier to obtain from oxides → therefore, sulphides and carbonates are first converted into oxides.

These ores are first converted into oxides because metal oxides are easier to reduce.



Roasting VS calcination

(Q) Differentiate between roasting and calcination with chemical equations.

Roasting	Calcination
Heating of <u>sulphide ores</u> in presence of <u>excess air (oxygen)</u> .	Heating of <u>carbonate ores</u> in the absence or limited supply of air .
Converts <u>sulphide ore</u> → oxide .	Converts <u>carbonate ore</u> → oxide .
Produces <u>SO₂ gas (sulphur dioxide)</u> , which is harmful and causes air pollution / acid rain .	Produces <u>CO₂ gas</u> , which is a greenhouse gas and contributes to global warming.
Example: $2\text{ZnS} + 3\text{O}_2 \rightarrow 2\text{ZnO} + 2\text{SO}_2$	Example: $\text{ZnCO}_3 \xrightarrow{\Delta} \text{ZnO} + \text{CO}_2$
<u>Zinc blende</u> →	Calamine →

Metal oxides are converted into Metals by reducing agents like Carbon.

For example: Heating zinc oxide with carbon yields metallic zinc.

zincite



↓
Reducing Agent



Reduction by Displacement (using highly reactive metals like Na, Ca, Al)

For example: Manganese dioxide reacts with aluminium powder.



Thermite reaction

A thermite reaction is a highly exothermic reaction in which aluminum powder reduces iron(III) oxide to molten iron.

Iron (III)
oxide
 Fe_2O_3

+

Aluminium
powder
 2Al



final product
 $2\text{Fe} + \text{Al}_2\text{O}_3 + \text{Heat}$



remember

Key points

- Highly exothermic: releases a large amount of heat.
- Produces iron in the molten state.
- Aluminium reduces iron(III) oxide to iron.



Uses

- Joining railway tracks
- Repairing cracked machine parts



Check yourself

QUES: Aluminium powder is used in thermit welding because:

- (a) Its reaction with iron is highly exothermic.
- ☒ (b) When it is heated with iron (III) oxide, molten iron is obtained.
- (c) When it is heated with iron (III) oxide, molten aluminium oxide is obtained to join railway tracks.
- (d) Its melting point is low as compared to iron and a molten alloy of iron and aluminium is formed on heating which is used to join railway tracks

Check yourself

QUES: A metal 'X' is used in the thermit process. When 'X' is heated with oxygen, it gives an oxide 'Y', which is amphoteric in nature. 'X' and 'Y' respectively are:

- (a) Mn, MnO_2
- ☒ (b) Al, Al_2O_3
- (c) Fe, Fe_2O_3
- (d) Mg, MgO

→ Al

↓

Al_2O_3



Check yourself

Ques: Compound 'X' reacts with aluminium and the reaction is used for joining railway tracks. $\rightarrow \text{Fe}_2\text{O}_3$

(a) Identify compound 'X'.

(b) Name the reaction involved.

(c) Write the balanced chemical equation for the reaction.

(d) State whether compound 'X' is oxidised or reduced during the reaction.



Check yourself

QUES. A railway engineer selects Aluminium powder instead of carbon powder to obtain molten iron from Iron (III) oxide during emergency rail-track repair.

Explain:

1. Why Aluminium is preferred
2. Identify the oxidizing and reducing agents
3. State one advantage of the reaction.

Answer:

1. Aluminum is more reactive than iron and has a greater affinity for oxygen.

OA Therefore, it removes oxygen from iron(III) oxide more effectively than

R carbon

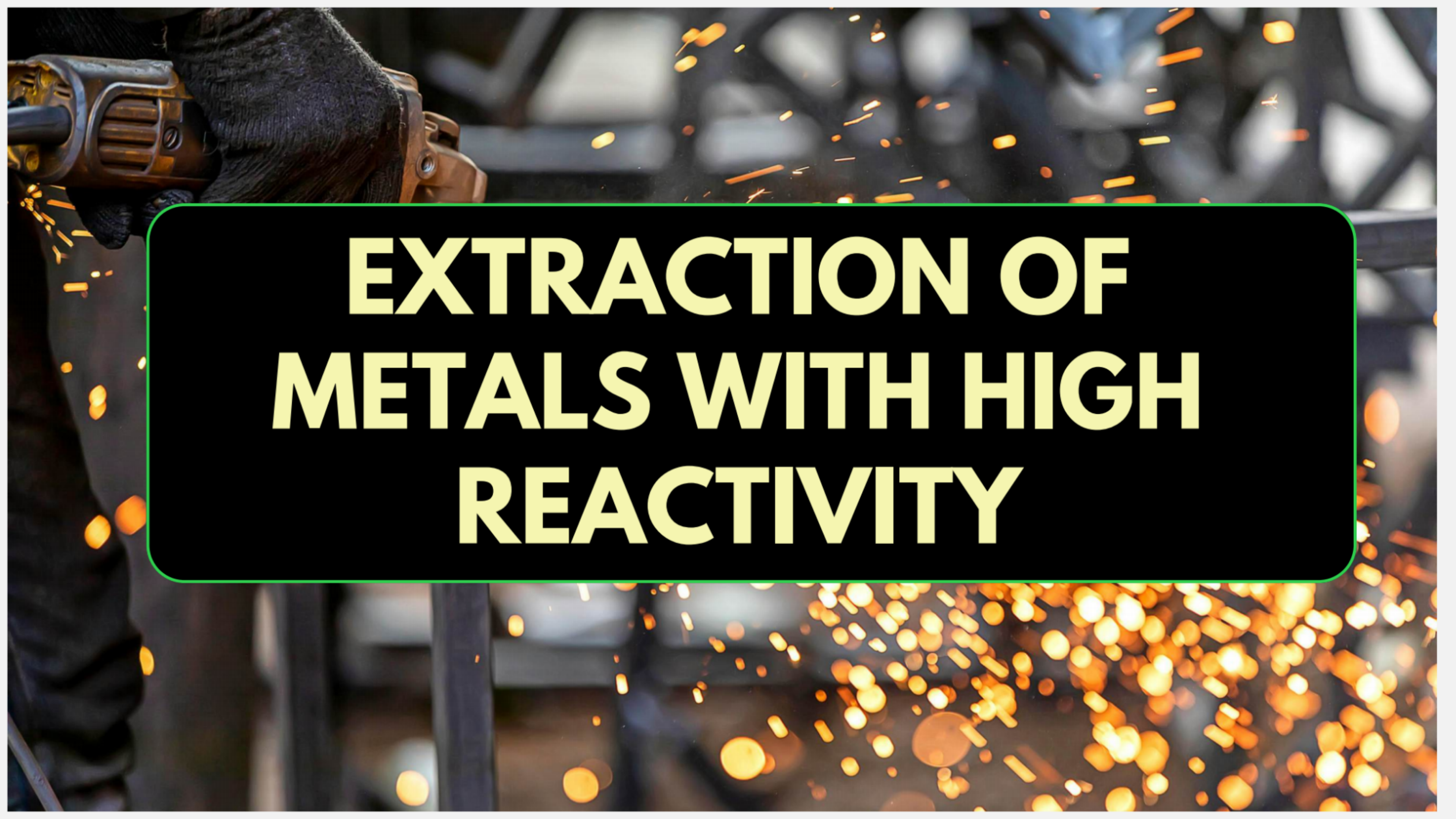


Oxidizing agent iron(III) oxide (Fe_2O_3) because it supplies oxygen.

Reducing agent: Al, because it removes oxygen from Fe_2O_3

3. The thermite reaction is highly exothermic, producing molten iron without an external furnace. This reaction makes it ideal for emergency welding and repairing railway tracks



A close-up photograph of a welding torch in operation. Bright orange and yellow sparks are flying out from the point of contact, creating a dynamic and industrial scene. The background is blurred, showing more of the industrial environment. Overlaid on the center of the image is a black rectangular box with rounded corners and a thin green border. Inside this box, the text "EXTRACTION OF METALS WITH HIGH REACTIVITY" is written in a bold, yellow, sans-serif font, arranged in three lines.

EXTRACTION OF METALS WITH HIGH REACTIVITY

Extraction of High reactive Metals

The molten compound is electrolyzed to obtain the pure metal.

- Potassium (K)
- Sodium (Na)
- Calcium (Ca)
- Magnesium (Mg)
- Aluminum (Al)
- *Sodium metal is deposited at the cathode.*
- *Chlorine gas is released at the anode*

